

Quiz 2 Solution

1. What will be the output of the following code snippet?

```
class A {
public:
    A() { cout << "Constructor A "; }
    ~A() { cout << "Destructor A "; }
};
class B {
public:
    A a;
    B() { cout << "Constructor B "; }
    ~B() { cout << "Destructor B "; }
};
int main() {
    B b;
    return 0;
}
```

- a) Constructor A Constructor B Destructor B Destructor A
- b) Constructor B Constructor A Destructor A Destructor B
- c) Constructor A Destructor A Constructor B Destructor B
- d) Destructor A Constructor A Destructor B Constructor B

Correct Answer: a) Constructor A Constructor B Destructor B Destructor A

2. What will happen if `delete` is called on a pointer that was not allocated using `new`?

- a) None of the others
- b) Undefined behavior may occur
- c) The memory is safely deallocated
- d) The compiler automatically corrects the error

Correct Answer: b) Undefined behavior may occur

3. Which of the following cannot be overloaded in C++?

- a) `[]`
- b) `sizeof`
- c) `++`
- d) `--`

Correct Answer: b) `sizeof`

4. Which of the following correctly overloads the `+=` operator for a class `MyClass` ?

- a) `MyClass& operator+=(const MyClass &obj);`
- b) `void operator+=(MyClass obj);`
- c) `void operator+=(MyClass &obj);`
- d) `friend MyClass operator+=(MyClass obj);`

Correct Answer: a) `MyClass& operator+=(const MyClass &obj);`

5. What is the time complexity of inserting an element at the end of a singly linked list?

- a) $O(1)$
- b) $O(\log n)$
- c) $O(n)$
- d) $O(n^2)$

Correct Answer: c) $O(n)$

6. If a class has a private member and a friend class, what happens?

- a) The friend class can access members of the class
- b) The friend class cannot create objects of the class
- c) The program will throw an error
- d) None of the others

Correct Answer: a) The friend class can create objects of the class

7. If a for loop (`for(int i=0, i<n;i++)`) runs while less than some number “n”, how many times will the variables i and n be compared?

- a) $n - 1$ times
- b) n times
- c) $n + 1$ times
- d) $n*2$ times

Correct Answer: c) $n + 1$ times

8. Which of the following operator overloads is NOT allowed in C++?

- a) Overloading the assignment operator (`=`)
- b) Overloading the subscript operator (`[]`)
- c) Overloading the member selection operator (`.`)
- d) Overloading the function call operator (`()`)

Correct Answer: c) Overloading the member selection operator (`.`)

9. Which of the following best describes the advantage of using a circular linked list over a singly linked list for queue implementation?

- a) It allows insertion and deletion in $O(1)$ time without needing a tail pointer
- b) It makes traversing the list easier
- c) We can dequeue the last enqueued element easily
- d) It allows random access to any element

Correct Answer: a) It allows insertion and deletion in $O(1)$ time without needing a tail pointer

10. Which of the following best describes the time complexity of accessing a random element in an unordered doubly linked list?

- a) $O(1)$
- b) $O(\log n)$
- c) $O(n)$
- d) $O(n \log n)$

Correct Answer: c) $O(n)$

Q11. Which one of these is an application of a Doubly Linked List?

- a) Storing a transaction log for a bank
- b) Storing hierarchical data like a tree structure
- c) Implementing direct access to elements in constant time $O(1)$
- d) Navigating through the web browser's history

Correct: d).

Q12. In programming expressions can be written in many formats, two such formats are *Infix* and *Postfix*.

Infix (operators are written normally): $6 * 3 / 9$

Postfix (operators are written at last): $6 3 * 9 /$

Which of the following is a correct evaluation of the Postfix expression $3 2 + 3 -$ using a stack?

- a) Push 3 $\rightarrow$ [3], Push 2 $\rightarrow$ [3, 2], Pop 3 & 2, compute $3 + 2 = 5$, Push 5 $\rightarrow$ [5], Push 3 $\rightarrow$ [5, 3], Pop 5 & 3, compute $5 - 3 = 2$, final result: 2
- b) Push 3 $\rightarrow$ [3], Push 2 $\rightarrow$ [2, 3], Pop 2 & 3, compute $2 + 3 = 5$, Push 5 $\rightarrow$ [5], Push 3 $\rightarrow$ [3, 5], Pop 3 & 5, compute $3 - 5 = -2$, final result: -2
- c) Push 2 $\rightarrow$ [2], Push 3 $\rightarrow$ [2, 3], Pop 2 & 3, compute $2 + 3 = 5$, Push 5 $\rightarrow$ [5], Push 3 $\rightarrow$ [5, 3], Pop 5 & 3, compute $5 - 3 = 2$, final result: 2
- d) Push 2 $\rightarrow$ [2], Push 3 $\rightarrow$ [3, 2], Pop 3 & 2, compute $3 + 2 = 5$, Push 5 $\rightarrow$ [5], Push 3 $\rightarrow$ [3, 5], Pop 3 & 5, compute $3 - 5 = -2$, final result: -2

Correct: a)

Q13. Which one of these is NOT a valid way to insert in a circular linked list?

```
void insert(int val) {
    Node* newNode = new Node(val);
    if (!last) {
        last = newNode;
        last->next = last;
    }
    else {
        newNode->next = last->next;
        last->next = newNode;
    }
}
```

a)

```
void insert(int val) {
    Node* newNode = new Node(val);
    if (!last) {
        last = newNode;
        last->next = last;
    }
    else {
        newNode->next = last->next;
        last->next = newNode;
        last = newNode;
    }
}
```

b)

```
void insert(int val) {
    Node* newNode = new Node(val);

    if (!last) {
        last = newNode;
        last->next = last;
        return;
    }

    Node* first = last->next;
    last->next = newNode;
    newNode->next = first;
    last = newNode;
}
```

c)

- a) b
- b) c
- c) a
- d) none of the others

Correct: d)

Q14. What is the array access cost in the following:

```
for(int i=0; !(a[i] == 1 || i>=N); i++){
    if(a[i] + b[i] == 0)
        cout<< "good luck for the quiz!"
}
```

- a) 0 to 3n
- b) 0 to 2n+1
- c) 1 to 2n
- d) 1 to 3n+1

Correct: d)

Q15. What is the Big O running time for the following:

```
while (N > 0) {  
    N /= 10;  
}
```

- a) $\text{Log}_2(N)$
- b) $\text{Log}_2(N/10)$
- c) $\text{Log}_{10}(N)$
- d) $\text{Log}_{10}(N/2)$

Correct: c)

Q16. Which of the following is FALSE about the order of growth?

- a) $N \log N$ is more expensive than N
- b) N^3 is less expensive than $N^2 \log N$
- c) N^2 is more expensive than $N \log N$
- d) N^3 is less expensive than 2^N

Correct: b)

Q17. Which of the following can be implemented using Linked Lists?

- a) Stack only
- b) Queue only
- c) Both
- d) None

Correct: c)

Q18. Resizable Arrays are superior over Linkedlists for:

- a) They allow elements to be directly accessed faster than linked lists.
- b) They always have a lower insertion time complexity than linked lists.
- c) They support constant-time deletion from any position.
- d) They occupy only as much memory as the size of the data.

Correct: a)

Q19. Insertions at the front or rear, in a Doubly LinkedList can be performed in:

- a) $O(N)$**
- b) $O(\log N)$**
- c) $O(1)$**
- d) None of the others.**

Correct: c)

Q20. Which of the following is TRUE about the code given below:

```
class Jhon {  
    private:  
        string Hisname;  
        int Hisage;  
  
    public:  
        Jhon() : Hisname("Jhon"), Hisage(20) {}  
  
        friend class Emily;  
};  
  
class Emily {  
    string Hername;  
    int Herage;  
  
    public:  
        Emily() : Hername("Emily"), Herage(21) {}  
  
};
```

- a) Emily can access all properties of Jhon.
- b) Jhon can access all properties of Emily.
- c) Both classes can access all properties of each other.
- d) None of the others.

Correct: a)